Structured Anchors Predict, Random Probes Calibrate: A Per-Image Conformal Layer for Sparse-Guided Depth

Supplementary Material

References to sections, equations, theorems and tables of the main paper use the main paper's numbering ("main Thm. 1"). All notation is as in main Sec. 3–4.

A. Proof of Theorem 1 (dual-role validity)

Setting. Fix an image x with its finite set $\mathcal{P}$ of valid pixels, its prediction $\hat{y}$, a structured anchor set $\mathcal{S} \subset \mathcal{P}$ and the values $\tilde{d}|_{\mathcal{S}}$ reported at $\mathcal{S}$. Everything below is conditional on $(x, \hat{y}, \mathcal{S}, \tilde{d}|_{\mathcal{S}})$; since the statement holds for every realization, it also holds after averaging over any randomness in the layout. The centre m, the scale $s > 0$, the per-image median $M = \mathrm{med}_{p\in\mathcal{P}}\, s(p)$ and hence $s_\gamma = M(s/M)^\gamma$ are, by assumption, functions of $(x, \hat{y}, \mathcal{S}, \tilde{d}|_{\mathcal{S}})$ and fixed constants (γ included); so they are fixed functions on $\mathcal{U} = \mathcal{P} \setminus \mathcal{S}$. Let $v : \mathcal{U} \to \mathbb{R}$ be the log residual of the quantity the probes report, defined at every pixel of $\mathcal{U}$ ($v = d$ for exact depth, $v = \tilde{d}$ for a sensor). Then every $p \in \mathcal{U}$ carries a fixed score

$$R(p) = \frac{|v(p) - m(p)|}{s_\gamma(p)}. \tag{1}$$

Let $N = |\mathcal{U}| \geq n + 1$, $r = \lceil (n + 1)(1 - \alpha) \rceil$.

Coverage event. Since $s_\gamma > 0$ and exp is increasing,

$$\begin{aligned} &\hat{y}(p)e^{v(p)} \in C(p) \\ &\iff m(p) - \hat{q}s_\gamma(p) \leq v(p) \leq m(p) + \hat{q}s_\gamma(p) \\ &\iff R(p) \leq \hat{q}. \end{aligned}$$

For $v = d$, $\hat{y}e^d = y$, so this is coverage of the true depth.

Exchangeability. Let $(Z_1, \ldots, Z_{n+1})$ be a uniformly random ordered sample without replacement from $\mathcal{U}$, and set $\mathcal{Q} = \{Z_1, \ldots, Z_n\}$ and $J = Z_{n+1}$. Then $\mathcal{Q}$ is a uniform n-subset of $\mathcal{U}$ and, given $\mathcal{Q}$, J is uniform on $\mathcal{U} \setminus \mathcal{Q}$, exactly as in the theorem. The law of $(Z_1, \ldots, Z_{n+1})$ is invariant under permutations, and R is a fixed function, so $(R(Z_1), \ldots, R(Z_{n+1}))$ is exchangeable.

Rank argument. If $r > n$, $\hat{q} = \infty$ and coverage is 1. Otherwise, break ties among the $n + 1$ scores uniformly at random and let ρ be the resulting rank of $R(Z_{n+1})$. By exchangeability ρ is uniform on $\{1, \ldots, n+1\}$. If $\rho \leq r$, at most $r-1$ of the n probe scores are ranked below $R(Z_{n+1})$, so at least $n - r + 1$ of them are $\geq R(Z_{n+1})$ and the r-th smallest probe score satisfies $\hat{q} \geq R(Z_{n+1})$. Hence

$$\mathbb{P}\big(R(J) \leq \hat{q}\big) \;\geq\; \mathbb{P}(\rho \leq r) \;=\; \frac{r}{n+1} \;\geq\; 1 - \alpha. \tag{2}$$

Without ties, $R(J) \leq \hat{q}$ holds if and only if $\rho \leq r$, so equality holds.

From one pixel to image coverage. Given $\mathcal{Q}$, J is uniform on $\mathcal{E} = \mathcal{U} \setminus \mathcal{Q}$, so $\mathbb{P}(R(J) \leq \hat{q} \mid \mathcal{Q}) = \mathrm{Cov}_i$ (main Eq. 2, with y replaced by $\hat{y}e^v$). Taking expectations over $\mathcal{Q}$, $\mathbb{E}[\mathrm{Cov}_i \mid x, \hat{y}, \mathcal{S}, \tilde{d}|_{\mathcal{S}}] = \mathbb{P}(R(J) \leq \hat{q}) \geq r/(n + 1)$. □

Remarks. (a) No assumption is made on how $\mathcal{S}$ was chosen, on the distribution of images, or on m and s beyond their independence of the probe values; this is why any layout and any probe-independent centre (*e.g.* a completion network fed RGB and $\mathcal{S}$) are allowed. (b) If any quantity used in m, s or γ were selected after looking at the probe values of the same image, R would no longer be a fixed function and the argument fails. Choosing γ once from aggregate results on NYU/KITTI is a milder form of the same issue; we therefore treat ARKitScenes as the only held-out check of efficiency (main Sec. 5). (c) The probes must be uniform over the pixels the guarantee concerns. If a sensor cannot return at some pixels (glass, sky, out of range), $\mathcal{U}$ is restricted to the pixels where it returns, and so is the guarantee. (d) This is the split-conformal argument for a fitting set and a calibration set inside one instance [3] and the within-field exchangeability argument [5], specialized to a frozen prior and a structured fitting set.

B. Proof of Proposition 1 (per-image coverage law)

Assume no ties among $\{R(p)\}_{p\in\mathcal{U}}$ and $r \leq n$. Sort $\mathcal{U}$ by score and let T be the position (in $1, \ldots, N$) of the r-th smallest probe, so $\hat{q}$ is the T-th smallest score in $\mathcal{U}$. Exactly $T - r$ non-probe pixels have score $\leq \hat{q}$, so $(N - n)\mathrm{Cov}_i = T - r$. For $T = r + j$, the sample must contain $r - 1$ of the first $r + j - 1$ pixels, the pixel at position $r + j$, and $n - r$ of the last $N - r - j$ pixels:

$$\mathbb{P}(T - r = j) = \frac{\binom{r+j-1}{r-1}\binom{N-r-j}{n-r}}{\binom{N}{n}}, \qquad j = 0, \ldots, N - n. \tag{3}$$

This negative-hypergeometric law equals $\mathrm{BetaBin}(N - n, r, n + 1 - r)$ (a standard identity, *e.g.* via the Pólya urn; we also checked it numerically to 10^{-14} for $N = 300$, $n = 20$). Its mean is $(N - n)\, r/(n + 1)$, so $\mathbb{E}[\mathrm{Cov}_i] = r/(n+1)$, the equality case of Theorem 1. A beta-binomial variable is a binomial mixture with success probability $B \sim$

$\mathrm{Beta}(r, n+1-r)$; dividing by $N-n$ and letting $N \to \infty$ with n fixed gives $\mathrm{Cov}_i \to B$ in distribution, the classical conditional-coverage law of split conformal prediction [7]. □

Numbers. Main Tab. 1 uses the Beta approximation. For $n = 20$ ($r = 19$): $\mathbb{E}[\mathrm{Cov}_i] = 19/21 = 0.905$; the 10th percentile is 0.819; $\mathbb{P}(\mathrm{Cov}_i < 0.85) = 17.6\%$ for one draw, the definition of short images throughout. With a few thousand evaluation pixels per image the finite-population correction is negligible at the reported precision. Matching these values empirically is an implementation check of the randomization, not a finding.

C. Proof of Theorem 2 (hull validity)

The per-bin guarantee of A-dist-split. Let scenes (rooms, drives, visits) be the exchangeable units: the U calibration scenes and one test scene are exchangeable, every image in them receives measurements by the same anchor rule (layout, k, n), and all hyper-parameters of m and s are fitted on separate head-train scenes. Within a scene, m, s and δ use the scene's own measurements, computed by the same rule in every scene; they are test-time features. For a distance bin b and threshold λ, let $L_{u,b}(\lambda)$ be the average over images of scene u (that have at least 10 evaluation pixels in bin b) of the fraction of their bin-b evaluation pixels with $|d-m|/s > \lambda$; it lies in $[0,1]$ and is non-increasing in λ. Scenes enter bin b by a rule applied identically to each scene and depending on pixel positions only, so the entering scenes remain exchangeable. Conformal risk control [1] with $\hat{\lambda}_b = \inf\{\lambda : \frac{U_b}{U_b+1}\hat{L}_b(\lambda) + \frac{1}{U_b+1} \le \alpha\}$, where $\hat{L}_b$ is the mean of $L_{u,b}$ over the U_b entering calibration scenes, gives $\mathbb{E}[L_{\mathrm{test},b}(\hat{\lambda}_b)] \le \alpha$: the expected bin-$b$ miscoverage of a new scene is at most α, for every bin (the Mondrian argument [8]). *Scope.* (i) Bins supported by fewer than $2/\alpha - 1$ calibration scenes are merged with the nearer neighbour; the guarantee then holds for the merged bin. Merging depends on calibration pixel positions only, never on residuals, but it is decided before the test scene is seen, a mild departure from fixing groups in advance. (ii) The guarantee is an average over scenes, not per image. (iii) Calibration and deployment must use the same anchor rule and sensor.

Hull. For every evaluation pixel, $C^{\mathrm{both}}(p) \supseteq C^{\mathrm{dual}}(p)$ and $C^{\mathrm{both}}(p) \supseteq C^{\mathrm{dist}}(p)$. Hence $\mathbf{1}\{y(p) \in C^{\mathrm{both}}(p)\} \ge \max\big(\mathbf{1}\{y(p) \in C^{\mathrm{dual}}(p)\}, \mathbf{1}\{y(p) \in C^{\mathrm{dist}}(p)\}\big)$ pointwise. Taking the per-image average and the expectation over probes gives Theorem 1's bound for A-dual-both; taking the bin-b average and the expectation over scenes gives the per-bin bound. That A-dist-split uses probe values in its own centre is irrelevant to the first statement, which only uses the A-dual interval. □

D. Sensor-error envelope (Proposition 2)

The main paper only points to this section. With iPad probes the envelope restored truth coverage (0.993–0.996) but over-covered: its interval score was 2–78% above that of A-dist-both (uniform 2–51%, clustered 60–78%). It thus failed the pre-specified gate (valid truth coverage and an interval score at least 10% below A-dist-both), so it is reported here.

With sensor-valued probes, Theorem 1 certifies the reading $\tilde{y}$, not the depth y. Let $e(p) = \log \tilde{y}(p) - \log y(p)$ and suppose a reference (here a laser scan) exists on calibration scenes. We (a) subtract a global bias $\hat{\beta}$, the median of e on head-train pixels, from the probe values, and (b) give each sensor-confidence class c (ARKit: 0/1/2) the band $[-a_c, b_c] = [Q_c(\lambda/2), Q_c(1-\lambda/2)]$ of the corrected head-train error, clipped to contain 0. One conformal risk control [1] over the U calibration scenes sets $\hat{\lambda}$: the largest grid value with $\frac{U}{U+1}\hat{L}(\lambda) + \frac{1}{U+1} \le \alpha_e$, $\hat{L}$ the mean fraction of a scene's pixels outside their band (none feasible: infinite bands). With $[\ell, u]$ the A-dual interval for the corrected reading at level $1-\alpha_s$, the enveloped interval is

$$C^{\mathrm{env}}(p) = \big[\ell(p)\, e^{-b_{c(p)}},\ u(p)\, e^{a_{c(p)}}\big]. \tag{4}$$

Proposition 2 (Envelope)**.** *If the sensor reading and its confidence are defined at every evaluation pixel, the miscoverage of y by C^{env}, averaged over test scenes and probe draws, is at most $\alpha_s + \alpha_e$.*

The per-image guarantee on y is not retained: the envelope's part holds on average over scenes. We fix $\alpha_s = \alpha_e = 0.05$ before the run, so that $\alpha_s + \alpha_e = \alpha$.

Proof of Proposition 2. Let $\tilde{y}$ be the sensor reading, $e = \log \tilde{y} - \log y$, $\hat{\beta}$ the bias estimated on head-train pixels, $\tilde{y}' = \tilde{y}e^{-\hat{\beta}}$ the corrected reading and $e' = e - \hat{\beta}$, so $y = \tilde{y}' e^{-e'}$. Run A-dual at level $1-\alpha_s$ with probes reporting $\tilde{y}'$; by Theorem 1 with $v = \log \tilde{y}' - \log \hat{y}$, for every test image $\mathbb{E}_{\mathcal{Q}}[\text{fraction of } \mathcal{E}_i \text{ with } \tilde{y}' \notin [\ell, u]] \le \alpha_s$.

The envelope term. The bands $[-a_c(\lambda), b_c(\lambda)] = [Q_c(\lambda/2), Q_c(1-\lambda/2)]$, clipped to contain 0, are quantiles of e' on head-train pixels only, so they form a fixed family, nested and shrinking as λ grows ($\lambda = 0$: infinite bands). Hence the loss $L_u(\lambda)$, the fraction of scene u's pixels whose e' lies outside the band of their class, is in $[0,1]$, non-decreasing in λ and $L_u(0) = 0$. The U calibration scenes and a test scene are exchangeable, and $\hat{\lambda}$ is the largest grid value (or 0) with $\frac{U}{U+1}\hat{L}(\lambda) + \frac{1}{U+1} \le \alpha_e$, $\hat{L}$ the mean of L_u over calibration scenes. This is conformal risk control [1] with loss bound 1, written for a loss that grows with λ (substitute $-\lambda$ for the usual non-increasing form); it gives $\mathbb{E}[L_{\mathrm{test}}(\hat{\lambda})] \le \alpha_e$: in expectation, at most α_e of a new scene's pixels have their corrected error outside their class

band. A single risk control over one class-adaptive parameter replaces separate thresholds per class because conformal risk control cannot certify a level below $1/(U+1)$, which is 0.026 with $U = 37$ calibration scenes per split; per-class one-sided targets proportional to the class shares (1.1, 4.3 and 94.6% of pixels for confidence 0, 1, 2) would be 0.0003, 0.001 and 0.024, and every threshold would be infinite.

Union bound. If $\tilde{y}'(p) \in [\ell(p), u(p)]$ and $e'(p) \in [-a_{c(p)}, b_{c(p)}]$, then $y(p) = \tilde{y}'(p)e^{-e'(p)} \in [\ell(p)e^{-b_{c(p)}}, u(p)e^{a_{c(p)}}] = C^{\text{env}}(p)$. So the fraction of a test scene's pixels that miss y is at most the fraction whose corrected reading misses $[\ell, u]$ plus the fraction whose corrected error leaves its band. The first is a pixel-weighted mean of per-image fractions, each with expectation at most α_s over the probes; the second has expectation at most α_e over scenes. Taking expectations bounds the truth miscoverage by $\alpha_s + \alpha_e$. □

The bound is an average over scenes: the envelope term has no per-image guarantee. It requires the sensor reading and its confidence at every evaluation pixel (true for ARKit's dense depth; for a sparse sensor, the population is restricted to pixels with returns). The levels are fixed before the run at $\alpha_s = \alpha_e = 0.05$; the grid has 400 log-spaced values in $[10^{-6}, 0.5]$.

Fitted envelope (E20). The fit uses only the sensor and the laser truth, so it is the same for every model and layout; over the 20 visit splits, $\hat{\lambda} = 0.016$–0.031. The high-confidence band is about $\pm 6\%$ ($a_2 = 0.052$–0.071, $b_2 = 0.048$–0.058 in log); the low- and medium-confidence bands are 0.28–0.53 in log. 2.4% of calibration pixels fall outside their band, at the risk-control limit $((U+1)\alpha_e - 1)/U$.

E. The distance-aware centre and scale

For pixel p, with anchor residuals $\tilde{d}$ at the anchor set $\mathcal{A}$ ($\mathcal{A} = \mathcal{S}$ for A-dual, all k measurements for A-dist-split):

- **Shift** $m(p)$, one of: *median* (image median of $\tilde{d}|_{\mathcal{A}}$); *knn* (median of the $K = 16$ nearest anchors); *kernel* (Gaussian-weighted mean of the 16 nearest, bandwidth $h \in \{0.05, 0.1, 0.2\}$ of the diagonal); *bilateral* (kernel weights times a Gaussian similarity in $\log \hat{y}$, bandwidth $h_d \in \{0.1, 0.3\}$).
- **Spread** $\sigma(p)$: median absolute deviation of the 16 nearest anchor residuals around their median (leave-one-out image spread for the median variant).
- **Scale** $s(p) = (\sigma(p) + \tau)\, g(\delta_{\mathcal{A}}(p))$, with floor τ, non-decreasing g from isotonic regression of normalized error on distance, capped at its 99.9th percentile on head-train.
- **Selection**: per anchor draw, the variant and bandwidths with the best held-out log interval score on 200 head-train images with the same layout ("auto"); τ, g and the cap per layout. No calibration or test data is used.

Distance bins: $\delta < 0.02$, 0.02–0.05, 0.05–0.1, 0.1–0.2, ≥ 0.2 of the image diagonal; a bin is merged into its nearer neighbour if fewer than 5% of head-train images have 10 pixels in it, or fewer than $2/\alpha - 1 = 19$ calibration scenes do. With the median variant, one bin and $g \equiv 1$, the model reduces to a single per-image shift and spread.

F. Which method gives which guarantee

Method	per image	per bin*	cal. scenes
A-local (structured)	uniform only	–	no
A-local-split	yes	–	no
A-dist-split	–	yes	yes
A-dist-both	yes	yes	yes
A-dual	yes	–	no
A-dual-both	yes	yes	yes
A-dual+env	sensor part	–	yes†
split-CP	–	–	yes

Table S1. Guarantees at level $1 - \alpha$. "Per image" is in expectation over the random probes and concerns the values the probes report. *Per distance bin, averaged over scenes. †With a reference sensor; the truth guarantee is a scene average. split-CP is marginal only.

G. Data and E19 implementation

ARKitScenes. Depth-upsampling validation split, 5,594 frames from 107 visits after dropping one video without confidence maps. Ground truth is the Faro laser scan rendered into each frame; the iPad Pro LiDAR depth is ARKit's fused LiDAR–RGB product with per-pixel confidence $\{0, 1, 2\}$ (94.6% of pixels at 2). Frames are rotated upright and reduced to 480×640 (ground truth: lower median of each 3×3 block; iPad: nearest low-resolution pixel); depth beyond 10 m is ignored. Valid pixels are kept on a stride-8 lattice (23.8 M pixels per model). At drawn anchors the iPad reads 1.6% nearer than the laser (mean signed log error).

Training data of the models. NYU and KITTI are not in the training mixtures of these models. UniDepthV2 [6]: 23 datasets, about 16 M images, including ARKitScenes (split not reported), ScanNet, ScanNet++, Taskonomy and HM3D; our frames are from the ARKitScenes validation split, so the domain was seen in training. Depth Anything V2 metric [9]: the indoor checkpoint is fine-tuned on Hypersim, the outdoor one on Virtual KITTI 2; ARKitScenes is not in its training data. Depth Pro [2]: 21 datasets; its first stage includes ARKitScenes, ScanNet and Hypersim, its second stage is synthetic only. This, besides the unreproduced pipeline, is why it is not used on ARKitScenes. Metric3D v2 [4]: 18 datasets (DDAD, Lyft, DrivingStereo, DIML, Argoverse2, Cityscapes, DSEC, Mapillary PSD,

Pandaset, UASOL, Virtual KITTI, Waymo, Matterport3D, Taskonomy, Replica, ScanNet, HM3D, Hypersim), without ARKitScenes. OMNI-DC [10]: five synthetic datasets only (Hypersim, IRS, TartanAir, BlendedMVS, Virtual KITTI), so it is zero-shot on ARKitScenes.

E19: completion centre. OMNI-DC [10] is run with the released v1.1 checkpoint and the settings of its demo script (three resolutions, median depth normalization, Depth Anything V2 prior), with no fine-tuning. Inputs are the RGB frame and a sparse depth map that holds a value only at the 180 structured anchors of the mask: the laser value ("clean") or the iPad value ("lidar"; anchors without an iPad return are left empty). The 20 probes and all other pixels are zero, so the network never sees a probe. Frames are zero-padded to a multiple of 16 and the output is cropped back; inference runs in fp32, as published. All variants of a frame (layouts $\times$ splits $\times$ value sources) run in one batch, so the RGB encoder and the prior are evaluated once per frame. Anchor and probe positions are exported from the calibration pipeline and checked to match its frames and pixel order exactly, so both centres are calibrated on identical positions, splits and evaluation pixels. No frame had a non-finite centre (5,594 kept). Calibration: split-CP takes the $\lceil (N+1)(1-\alpha) \rceil$-th smallest $|\log y - \log \hat{y}|$ over the N evaluation pixels of all calibration frames (laser y); A-dist-split, A-dual and A-dual-both are as in the main paper, with the distance-aware model refit around each centre (variant chosen on 200 head-train frames) and $\gamma = 0.5$. Seeds 0–9 give ten visit splits, each with its own anchor mask; one probe draw per split. With laser anchors (both probe sources) E19 is confirmatory: ten splits, analysed by the fixed plan (paired bootstrap over splits, 10^4 resamples). OMNI-DC completed from iPad anchors was not part of the confirmatory run by design and is not reported. Point accuracy on the test pixels outside the structured anchors (descriptive; image-mean AbsRel, mean over the ten splits, min–max over splits in brackets): OMNI-DC reaches 0.0064 (0.0061–0.0068) under uniform and 0.0738 (0.0709–0.0757) under clustered anchors. UniDepthV2 rescaled per image by the median $\log(y/\hat{y})$ at the same 180 anchors reaches 0.0293 (0.0276–0.0305) and 0.0394 (0.0370–0.0418); without anchors it reaches 0.121. $\delta < 1.25$: 0.997 vs. 0.988 (uniform), 0.944 vs. 0.981 (cluster). Under uniform anchors OMNI-DC is about $4.6\times$ more accurate; under clustered anchors one per-image scale on UniDepthV2 roughly halves its error, matching the interval-score result.

Interval-score blow-up. Only for OMNI-DC under clustered anchors, the arms built on the distance model (A-dist-split, A-dual-both, and A-dual in splits where its scale field falls back to the kernel form) have, in some splits, a mean-over-images relative interval score of up to 10^{29}. OMNI-DC reproduces its own input anchors, so the anchor residuals that the distance model reads are almost 0; its spread estimate and the floor τ collapse to about 10^{-6}, the per-bin threshold in far bins becomes huge, and threshold times scale puts upper bounds of e^{tens} on 0.1–0.4% of pixels (Unin. column). Median widths and the primary IS (0.31–0.42) are normal. This is a property of calibrating a centre that has consumed the anchors with anchor residuals; A-dual's probe-based scale is far less affected. The primary score of main Sec. 5 (per image, the mean over pixels of the interval score over the true depth; then the median over images) is robust to such images and keeps the miss penalty, whereas a per-pixel median would mostly ignore misses at 90% coverage (smoke test, A-dual: 0.047 pixel median vs. 0.088 image median of means vs. 0.103 plain mean).

H. E18: value-source factorial over 20 splits

UniDepthV2 and DA-V2 (indoor metric fine-tune), frozen, on all 5,594 ARKitScenes frames; $k = 200$ (180 structured anchors, 20 probes), $\alpha = 0.1$, $\gamma = 0.5$; uniform and cluster layouts; anchors and probes each report the laser or the iPad value; 20 visit splits (seeds) $\times$ 10 probe draws. Preprocessing, splits and the distance-aware model are as in Sec. G; split-CP and OMNI-DC are not part of E18. p10 is the 10th percentile of per-image coverage for one draw, and the short share the fraction of images below 0.85 on one draw (law: 17.6%); both are averaged over runs. Confirmatory, analysed by the fixed plan (paired bootstrap over splits, 10^4 resamples). Standard errors over splits of A-dual's mean coverage are about 10^{-4} where the probes report the laser value. The uniform-layout UniDepthV2 cells agree with E19 (iPad/iPad A-dual truth coverage 0.566 here, 0.565 over ten splits in E19). A-dist-both covers the truth in every cell (at least 0.92), but under uniform anchors at 1.8–2.3$\times$ A-dual's interval score with laser values, and with iPad values for both its per-image spread is no longer that of the law (UniDepthV2: p10 0.781). Under clustered anchors the two are close (ratio 1.0), with A-dist-both slightly ahead: A-dual's interval score is 2.3–3.4% higher in the cells with laser probes (paired intervals exclude 0). Depth Pro was run in the same grid but is excluded: ARKitScenes is in its training data, and its predictions are not yet reproduced with the official package (Sec. G). Figure S1 shows why the probe source decides truth coverage: the iPad reads a few per cent nearer than the laser over much of a frame, so probes that report the iPad calibrate intervals around the iPad's depth, and the laser truth falls beyond them.

I. Additional results

Table S5 gives the paired bootstrap intervals behind the interval-score comparisons of main Sec. 6.2, and Tab. S6 the coverage per distance bin behind main Fig. 3a for every

			Uniform layout						Cluster layout					
Centre	Anch./probes	Calib.	Cov.	p10	Sh. %	W/y	IS	Unin. %	Cov.	p10	Sh. %	W/y	IS	Unin. %
UniDepthV2	laser / laser	split-CP	0.897	0.594	14.5	0.513	0.528	0.0	0.897	0.595	14.6	0.513	0.529	0.0
		A-dist-split	0.923	0.860	7.6	0.027	0.082	0.0	0.924	0.837	12.5	0.071	0.135	0.0
		A-dual	0.904	0.818	17.7	0.030	0.089	0.0	0.905	0.819	17.6	0.109	0.182	0.0
		A-dual-both	0.949	0.905	0.8	0.038	0.087	0.0	0.967	0.930	0.5	0.126	0.174	0.0
	iPad / laser	split-CP	0.897	0.594	14.5	0.513	0.528	0.0	0.897	0.595	14.6	0.513	0.529	0.0
		A-dist-split	0.616	0.390	93.7	0.035	0.144	0.0	0.942	0.868	7.0	0.098	0.152	0.0
		A-dual	0.904	0.818	17.9	0.066	0.121	0.0	0.906	0.820	17.5	0.123	0.191	0.0
		A-dual-both	0.912	0.836	13.4	0.068	0.120	0.0	0.972	0.938	0.3	0.146	0.188	0.0
	laser / iPad	split-CP	0.897	0.594	14.5	0.513	0.528	0.0	0.897	0.595	14.6	0.513	0.529	0.0
		A-dist-split	0.924	0.864	6.4	0.030	0.083	0.0	0.925	0.840	12.0	0.090	0.152	0.0
		A-dual	0.973	0.942	0.1	0.074	0.112	0.0	0.927	0.823	13.3	0.131	0.197	0.0
		A-dual-both	0.975	0.946	0.0	0.074	0.113	0.0	0.971	0.932	0.9	0.148	0.193	0.0
	iPad / iPad	split-CP	0.897	0.594	14.5	0.513	0.528	0.0	0.897	0.595	14.6	0.513	0.529	0.0
		A-dist-split	0.527	0.284	97.0	0.033	0.168	0.0	0.866	0.746	33.0	0.081	0.157	0.0
		A-dual	0.565	0.108	73.8	0.036	0.174	0.0	0.848	0.623	33.6	0.119	0.209	0.0
		A-dual-both	0.666	0.326	70.2	0.045	0.150	0.0	0.946	0.882	6.0	0.137	0.189	0.0
OMNI-DC	laser / laser	split-CP	0.897	0.813	20.7	0.024	0.073	0.0	0.895	0.676	23.7	0.336	0.374	0.0
		A-dist-split	0.923	0.850	9.9	0.026	0.071	0.0	0.924	0.757	16.7	0.391	0.422	0.2
		A-dual	0.904	0.819	17.8	0.023	0.074	0.0	0.905	0.820	17.4	0.226	0.311	0.0
		A-dual-both	0.948	0.902	1.3	0.032	0.074	0.0	0.971	0.917	2.2	0.403	0.421	0.2
	laser / iPad	split-CP	0.897	0.813	20.7	0.024	0.073	0.0	0.895	0.676	23.7	0.336	0.374	0.0
		A-dist-split	0.923	0.851	9.9	0.026	0.071	0.0	0.924	0.757	16.7	0.391	0.422	0.2
		A-dual	0.980	0.956	0.0	0.075	0.108	0.0	0.915	0.807	17.4	0.241	0.322	0.0
		A-dual-both	0.980	0.957	0.0	0.075	0.108	0.0	0.971	0.912	3.1	0.403	0.423	0.2

Table S2. **Full E19 grid** (ARKitScenes, $k = 200$, $n = 20$, $\alpha = 0.1$; mean over ten visit splits, one probe draw; confirmatory; –: IS dominated by the blow-up below). Anchors / probes: the value source of the 180 structured anchors (also OMNI-DC's input) and of the 20 probes. Coverage is of the laser truth. For UniDepthV2, split-CP uses the raw prediction (no anchors), so its rows repeat; for OMNI-DC it depends on the anchor source only. Columns as in main Tab. 3 (Sh.: short); Unin.: % of pixels with an interval wider than $10y$ (infinite included).

layout.

References

[1] Anastasios N. Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. Conformal risk control. In *International Conference on Learning Representations (ICLR)*, 2024. 2

[2] Aleksei Bochkovskii, Amaël Delaunoy, Hugo Germain, Marcel Santos, Yichao Zhou, Stephan R. Richter, and Vladlen Koltun. Depth pro: Sharp monocular metric depth in less than a second. In *International Conference on Learning Representations (ICLR)*, 2025. 3

[3] Yu Gui, Rina Foygel Barber, and Cong Ma. Conformalized matrix completion. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 1

[4] Mu Hu, Wei Yin, Chi Zhang, Zhipeng Cai, Xiaoxiao Long, Hao Chen, Kaixuan Wang, Gang Yu, Chunhua Shen, and Shaojie Shen. Metric3D v2: A versatile monocular geometric foundation model for zero-shot metric depth and surface normal estimation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(12):10579–10596, 2024. 3

[5] Huiying Mao, Ryan Martin, and Brian J. Reich. Valid model-free spatial prediction. *Journal of the American Statistical Association*, 119(546):904–914, 2024. 1

[6] Luigi Piccinelli, Christos Sakaridis, Yung-Hsu Yang, Mattia Segu, Siyuan Li, Wim Abbeloos, and Luc Van Gool. UniDepthV2: Universal monocular metric depth estimation made simpler. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pages 1–14, 2025. 3

[7] Vladimir Vovk. Conditional validity of inductive conformal predictors. In *Proceedings of the Asian Conference on Machine Learning (ACML)*, pages 475–490, 2012. 2

[8] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005. 2

[9] Lihe Yang, Bingyi Kang, Zilong Huang, Zhen Zhao, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. Depth anything V2. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. 3

[10] Yiming Zuo, Willow Yang, Zeyu Ma, and Jia Deng. OMNI-DC: Highly robust depth completion with multiresolution depth integration. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9287–9297, 2025. 4

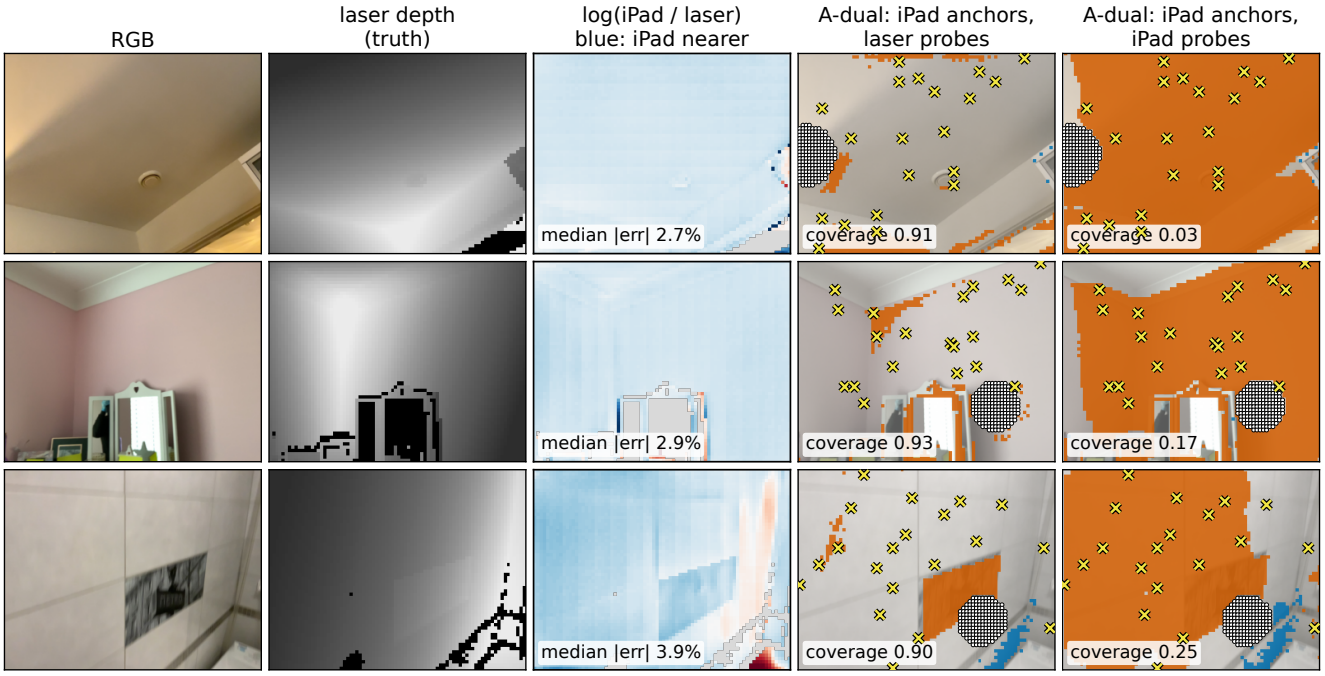


Figure S1. **Probe source and truth coverage** (three ARKitScenes test frames from different visits; UniDepthV2, clustered layout, 180 structured anchors (•, one cluster) reporting the iPad value, 20 probes (×); split 0, one draw). Third column: log ratio of the iPad LiDAR to the laser depth (blue: iPad nearer; grey: no value). Right two columns: A-dual's misses of the laser truth on the evaluation pixels (vermillion: truth farther than the interval, blue: nearer; dark grey: no ground truth), with laser and with iPad probes. Frames chosen among those whose coverage drops most from laser to iPad probes while staying near 0.9 with laser probes; averages over all frames are in Tab. S3.

			Uniform layout				Cluster layout			
Model	Anchors / probes	Calibration	Cov.	p10	Short %	IS	Cov.	p10	Short %	IS
UniDepthV2	laser / laser	A-local-split	0.909	0.794	19.7	0.189	0.909	0.795	19.7	0.189
		A-dist-split	0.924	0.862	7.1	0.081	0.924	0.838	12.4	0.133
		A-dist-both	0.979	0.953	0.0	0.157	0.974	0.939	0.5	0.177
		A-dual	0.905	0.819	17.6	0.088	0.905	0.819	17.7	0.181
		A-dual-both	0.949	0.906	0.7	0.086	0.968	0.931	0.5	0.173
	iPad / laser	A-local-split	0.909	0.794	19.7	0.189	0.909	0.795	19.7	0.189
		A-dist-split	0.617	0.389	93.4	0.144	0.942	0.869	7.1	0.152
		A-dist-both	0.968	0.926	1.4	0.164	0.978	0.949	0.3	0.185
		A-dual	0.905	0.819	17.5	0.120	0.905	0.819	17.6	0.190
		A-dual-both	0.913	0.837	13.2	0.119	0.972	0.937	0.4	0.188
	laser / iPad	A-local-split	0.867	0.656	30.4	0.212	0.867	0.656	30.4	0.212
		A-dist-split	0.925	0.867	5.9	0.083	0.926	0.841	11.5	0.152
		A-dist-both	0.978	0.952	0.2	0.166	0.970	0.930	1.5	0.192
		A-dual	0.973	0.942	0.2	0.112	0.926	0.822	13.6	0.197
		A-dual-both	0.975	0.946	0.1	0.112	0.971	0.932	1.0	0.193
	iPad / iPad	A-local-split	0.867	0.656	30.4	0.212	0.867	0.656	30.4	0.212
		A-dist-split	0.527	0.283	97.0	0.167	0.867	0.749	32.4	0.156
		A-dist-both	0.920	0.781	15.9	0.180	0.958	0.905	3.8	0.191
		A-dual	0.566	0.111	74.0	0.172	0.848	0.624	33.7	0.209
		A-dual-both	0.667	0.327	70.4	0.149	0.946	0.883	5.8	0.189
DA-V2	laser / laser	A-local-split	0.909	0.794	19.8	0.269	0.909	0.794	19.7	0.269
		A-dist-split	0.924	0.859	7.9	0.089	0.925	0.837	12.4	0.168
		A-dist-both	0.981	0.958	0.0	0.225	0.977	0.943	0.5	0.250
		A-dual	0.905	0.819	17.6	0.096	0.905	0.819	17.5	0.257
		A-dual-both	0.949	0.906	0.8	0.094	0.971	0.933	0.6	0.244
	iPad / laser	A-local-split	0.909	0.794	19.8	0.269	0.909	0.794	19.7	0.269
		A-dist-split	0.696	0.499	86.6	0.136	0.938	0.860	8.4	0.181
		A-dist-both	0.973	0.936	0.8	0.230	0.980	0.950	0.4	0.254
		A-dual	0.905	0.819	17.5	0.125	0.905	0.819	17.6	0.261
		A-dual-both	0.916	0.846	11.1	0.124	0.973	0.937	0.5	0.252
	laser / iPad	A-local-split	0.890	0.731	26.1	0.283	0.890	0.729	26.2	0.283
		A-dist-split	0.924	0.862	7.2	0.090	0.926	0.840	11.8	0.182
		A-dist-both	0.981	0.956	0.1	0.232	0.975	0.938	1.0	0.260
		A-dual	0.969	0.933	0.4	0.116	0.920	0.812	15.8	0.269
		A-dual-both	0.972	0.940	0.1	0.116	0.973	0.933	0.8	0.260
	iPad / iPad	A-local-split	0.890	0.731	26.1	0.283	0.890	0.729	26.2	0.283
		A-dist-split	0.620	0.391	93.8	0.154	0.891	0.786	23.8	0.184
		A-dist-both	0.945	0.865	8.5	0.241	0.969	0.927	1.8	0.260
		A-dual	0.632	0.180	68.3	0.166	0.877	0.722	27.9	0.274
		A-dual-both	0.740	0.451	62.0	0.143	0.960	0.909	2.8	0.255

Table S3. **Full E18 grid** (ARKitScenes, frozen models, $k = 200$, $n = 20$, $\alpha = 0.1$; mean over 20 visit splits $\times$ 10 probe draws; confirmatory). Coverage is of the laser truth. p10: one draw; short: % of images whose coverage on one draw is below 0.85; IS: relative interval score (median over images).

		Coverage		Short (%)	Δ width (%)		Δ IS (%)	
Layout	k	local	dual	dual	dual	both	dual	both
uniform	50	[0.923, 0.923]	[0.905, 0.905]	[17.3, 17.6]	[−52, −44]	[−42, −22]	[−23, −9]	[−27, −11]
	200	[0.901, 0.901]	[0.904, 0.905]	[17.2, 17.9]	[−72, −60]	[−66, −45]	[−42, −21]	[−42, −23]
rows	50	[0.893, 0.901]	[0.905, 0.905]	[17.4, 17.8]	[−45, −38]	[−34, −18]	[−17, −7]	[−22, −9]
	200	[0.869, 0.876]	[0.905, 0.905]	[17.3, 17.8]	[−58, −48]	[−46, −31]	[−29, −13]	[−30, −15]
cluster	50	[0.305, 0.417]	[0.905, 0.905]	[17.4, 17.8]	[−24, −11]	[−3, 0]	[−1, 5]	[−2, 1]
	200	[0.371, 0.469]	[0.905, 0.905]	[17.5, 17.7]	[−25, −12]	[−5, −1]	[−1, 5]	[−2, 1]
edges	50	[0.941, 0.967]	[0.904, 0.905]	[17.6, 17.8]	[−28, −16]	[−16, −4]	[−7, 3]	[−11, 0]
	200	[0.924, 0.960]	[0.905, 0.905]	[17.5, 17.6]	[−38, −31]	[−30, −13]	[−16, −5]	[−20, −7]
near	50	[0.749, 0.801]	[0.904, 0.905]	[17.4, 17.7]	[−48, −20]	[−29, −11]	[−17, −1]	[−19, −7]
	200	[0.705, 0.760]	[0.904, 0.905]	[17.5, 17.8]	[−53, −21]	[−38, −12]	[−22, 4]	[−25, −4]

Table S4. **Ranges [min, max] over the six NYU/KITTI model–dataset pairs** for main Tab. 2 (DA-V2, UniDepthV2, Depth Pro; 20 scene splits × 10 probe draws; confirmatory). Δ against A-dist-both. Short: one draw.

		A-dual			A-dual-both		
Layout	k	Δ	95% CI	$-/+$	Δ	95% CI	$-/+$
uniform	50	-19.3	$[-23.6, -8.1]$	6/0	-22.7	$[-27.7, -10.4]$	6/0
	200	-37.2	$[-43.5, -20.3]$	6/0	-38.6	$[-43.4, -21.9]$	6/0
rows	50	-14.5	$[-18.1, -6.5]$	6/0	-18.5	$[-22.7, -9.0]$	6/0
	200	-21.5	$[-30.1, -11.9]$	6/0	-25.6	$[-31.3, -14.7]$	6/0
cluster	50	3.1	$[-1.3, 5.5]$	1/5	-0.4	$[-1.9, 1.4]$	3/2
	200	2.5	$[-1.7, 5.6]$	1/4	-1.3	$[-2.3, 1.0]$	4/2
edges	50	-4.1	$[-8.4, 4.0]$	5/1	-7.9	$[-11.3, 0.7]$	5/0
	200	-7.7	$[-17.7, -4.7]$	6/0	-12.3	$[-21.0, -6.2]$	6/0
near	50	-6.5	$[-17.8, -0.9]$	6/0	-9.7	$[-19.7, -6.1]$	6/0
	200	-5.6	$[-22.7, 5.1]$	3/3	-10.3	$[-25.1, -3.5]$	6/0

Table S5. **Relative interval score against A-dist-both** (E17c, NYU/KITTI, $n = 20$; 20 splits $\times$ 10 draws). Δ: median over the six pairs of the relative difference (%); 95% CI: lowest lower and highest upper bound over pairs of the paired bootstrap interval over splits (10^4 resamples), relative to A-dist-both; $-/+$: pairs whose interval lies entirely below / above 0.

Layout		<0.02	0.02–0.05	0.05–0.1	0.1–0.2	≥0.2
uniform	pixels (%)	24.6	46.9	25.1	3.4	0.0
	A-local	0.94	0.93	0.89	0.83	–
	A-dist-both	0.98	0.97	0.96	0.95	–
	A-dual	0.93	0.91	0.87	0.82	–
	A-dual-both	0.96	0.95	0.95	0.94	–
rows	pixels (%)	22.8	44.0	26.9	6.1	0.2
	A-local	0.93	0.91	0.87	0.81	0.69
	A-dist-both	0.97	0.97	0.96	0.95	0.92
	A-dual	0.93	0.91	0.88	0.84	0.74
	A-dual-both	0.96	0.96	0.95	0.94	0.91
cluster	pixels (%)	12.1	34.7	36.4	15.7	1.2
	A-local	0.78	0.63	0.51	0.42	0.30
	A-dist-both	0.98	0.97	0.97	0.96	0.94
	A-dual	0.93	0.92	0.90	0.86	0.80
	A-dual-both	0.97	0.97	0.96	0.95	0.94
edges	pixels (%)	20.6	41.2	30.4	7.4	0.3
	A-local	0.96	0.96	0.97	0.95	0.91
	A-dist-both	0.96	0.97	0.97	0.96	0.91
	A-dual	0.90	0.91	0.91	0.87	0.79
	A-dual-both	0.96	0.96	0.96	0.96	0.91
near	pixels (%)	21.2	41.9	27.8	8.6	0.5
	A-local	0.93	0.88	0.74	0.61	0.57
	A-dist-both	0.98	0.97	0.96	0.95	0.92
	A-dual	0.93	0.91	0.87	0.81	0.74
	A-dual-both	0.97	0.96	0.96	0.95	0.92

Table S6. **Coverage by distance to the nearest anchor** (fraction of the image diagonal; E17c, $k = 50$, $n = 20$, median over the six NYU/KITTI pairs). First row of each layout: median pixel share; –: bin holds less than 0.1% of pixels.